

SECONDARY DATA ANALYSIS

A REPORT ON ANALYSIS OF SECONDARY DATA ON INFLATION EXPECTATIONS COLLECTED FROM IESH

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An Insight on the Data:

The Secondary Data provided, has been collected from Inflation Expectations Survey of Households (IESH), which was introduced by the Reserve Bank of India (RBI) from September 2005. The primary purpose of the survey was to collect regional heterogeneity of inflation expectations of households. The data considered for our analysis considers inflation expectations of households from July 2008 to September 2019. Apart from the data collected from IESH, we have taken data on monthly inflation of India from The Labour Bureau, Ministry of Labour and Employment, Government of India, and have taken the mean of the inflation rates to take it down to round wise data. In this analysis, each of the actual inflation rates considered is a mean of 3 consecutive monthly inflation rate of the country.

Coverage of the data:

The survey was conducted in 12 cities. From each metropolitan city, 500 households and 250 households from each of the remaining 8 cities were selected in the sample, thus making the total sample size to 4000 households. The country is divided into 4 zones - North: Delhi, Jaipur, Lucknow, Bhopal; South: Chennai, Bangalore, Hyderabad; East: Kolkata, Guwahati, Patna; and West: Mumbai, Ahmedabad, to get an overall coverage. Respondents above 18 years of age are only considered.

Insight on the analysis:

This analysis seeks to answer key questions about the relationship between households' inflation expectations and the actual inflation rate in India. It explores how inflation expectations are influenced by various factors, such as geographical location, occupation, age, gender, and commodity type. It also investigates whether these expectations are aligned with the actual inflation rate observed over time and whether they can serve as reliable predictors for future inflation trends.

By examining these questions, the analysis aims to provide valuable insights into the public's perception of inflation, which is essential for policymakers, economists, and other stakeholders in understanding the dynamics of inflation expectations and their potential impact on economic decision-making.

Query 1: Can expected current inflation rate be treated as a predictor of actual inflation rate on the country?

For examining if expected current inflation can be treated as a predictor of actual inflation rate of the country, we consider a bivariate data (x_i, y_i) on 216160 individuals, where x_i and y_i respectively denotes the actual inflation rate and expected current inflation rate by respondents at the same time period.

We try to fit a linear regression taking the expected inflation as predictor and the actual inflation as response.

The best fitted line is given by:

Y = 6.141 + 0.135x, where Y is the predicted actual inflation based on the model. But this model gives the coefficient of determination as 0.03771, suggesting that the model explains only 3.77% of the total variation in actual inflation.

To further examine if the relation between predicted and response variables can be explained by any polynomial regression model, we fit a cubic model, which is given by,

y = 7.963 – 0.844x + 0.132 x^2 - 0.005 x^3

Here the correlation index of order 3 is obtained as 0.0538, suggesting that the cubic regression model explains only 5.38% of the total variation in actual inflation, and the remaining 94.62% remains unexplained. Further increasing the degree of the polynomial, we observe that R^2 increases by negligible quantity. Hence, we do not go for higher degree regression polynomial.

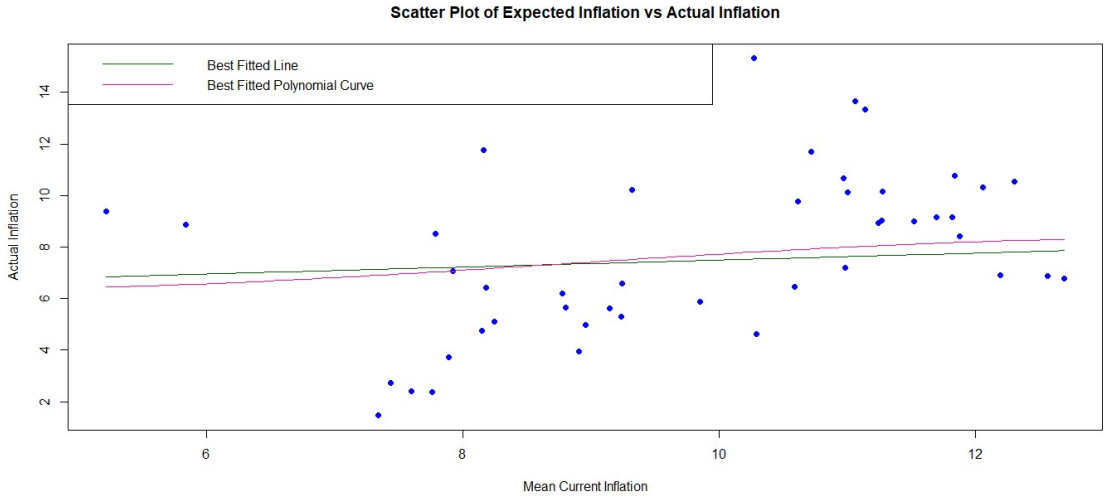


Fig 1

In the above scatterplot of Mean Current Expected Inflation and Actual inflation, we have plotted mean anticipated inflation along X-axis and actual inflation rate along Y-axis. Mean expected inflation rate is considered in order to have one representative value for each round, and since the data does not have any extreme values, mean is a suitable measure. Otherwise, plotting actual inflation vs expected current inflation, since actual inflation is same for multiple values of x (since it is same for all data corresponding to a single round), it becomes difficult

to draw visual conclusions. Here, visually also, it is evident that neither the linear nor the polynomial equation fits well to the data.

Hence, the expected inflation of the respondent is not at all a good predictor for actual inflation, if we consider linear regression (i.e., regression equation which is linear in parameters).

Query 2: How do expectations regarding current and future inflation (for the next year) differ across various age groups?

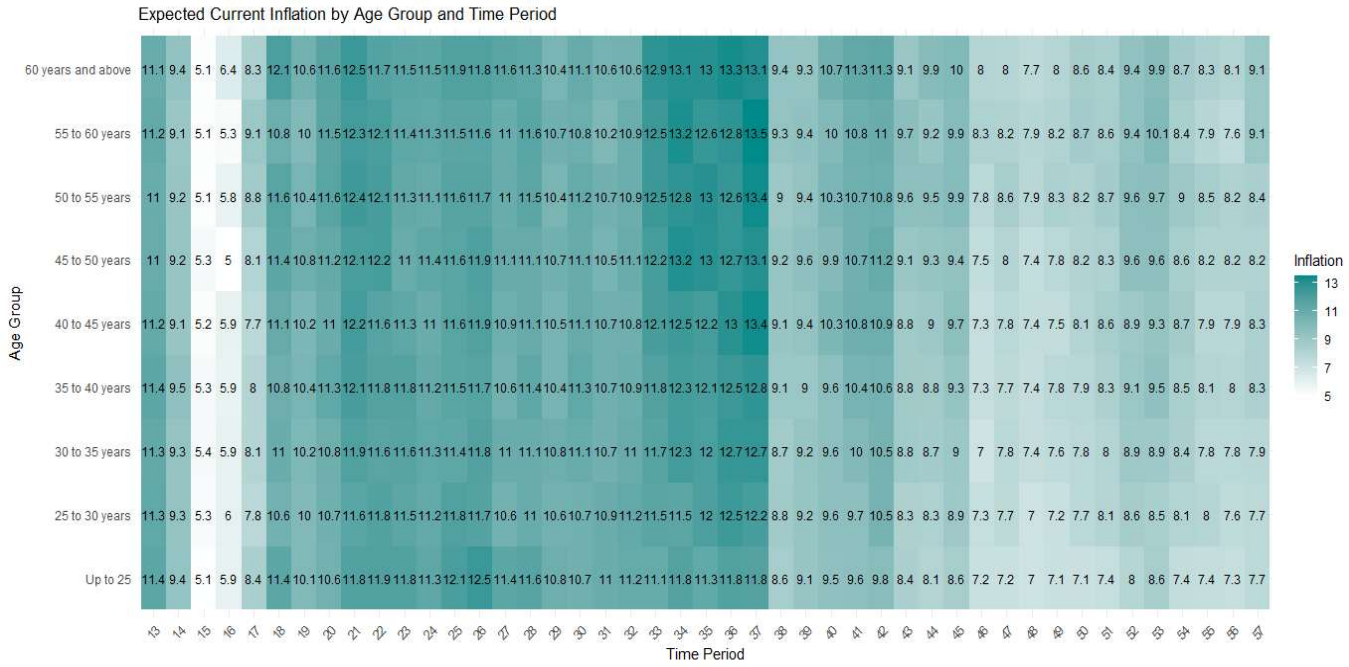


Fig 2

Expected current inflation rate by different age groups over time:

Heatmap is an effective way for visual representation of the mean inflation of age groups over the rounds. The above heatmap shows current expected inflation rate by different age groups over time. The X-axis plots the Round Number and the Y-axis is for the different age groups. Here each cell shows the mean expected current inflation rate for the corresponding round and age group. Dark coloured areas show high expected inflations, and lighter shades show lower expected inflation.

Analysis across Rounds: The heatmap analysis reveals a dynamic landscape of expected inflation across different age demographics over the time period July 2008 to Sept 2019. Notably, the data highlights a sharp decline in inflation expectations between rounds 15 and

16. The peak inflation expectations observed between rounds 33 to 37, coinciding with July 2013 to September 2014, suggest a period of heightened inflationary pressures. Following this peak, inflation expectations stabilized around 8-9%, indicating a potential anchoring of expectations or the effectiveness of implemented policies.

Analysis across Age Groups: Furthermore, the analysis demonstrates a consistent pattern where older age groups generally maintain higher inflation expectations compared to younger ones. Despite these differences in magnitude, all age groups exhibit a similar directional movement in their expectations.

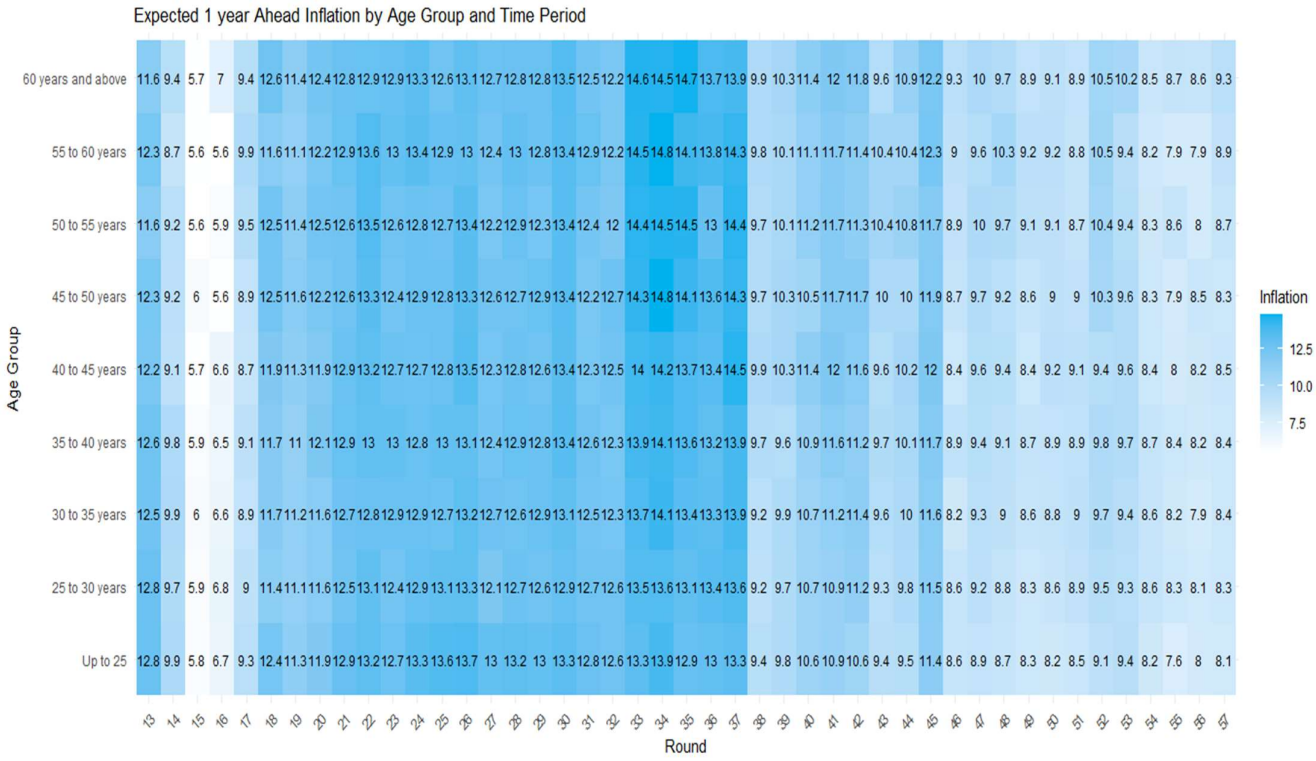


Fig 3

Expected 1 year ahead inflation rate by different age groups over time:

This heatmap shows 1 year ahead inflation rate expected by different age groups over time. In X-axis we plot the rounds and the Y-axis shows the different age groups.

The analysis of the heatmap depicting expected 1-year ahead inflation rates across nine age groups over 45 rounds reveals a dramatic drop from approximately 12.5-13.5% in round 13 to 5.5-7% in rounds 15 and 16 suggests a period of rapid adjustment, potentially influenced by policy shifts or sudden economic changes. Conversely, rounds 33 to 37 (July-September 2013 to July-September 2014) saw the highest expected inflation, reaching around 14%, indicating a period of heightened expected inflationary pressures. Following this peak, a substantial decrease to 9-10% occurred, with expectations stabilizing within the 8-10% range thereafter.

Across age groups, a consistent pattern emerges where respondents in older groups consistently report higher expected inflation rates compared to younger groups. However, the directional

alignment of expectations across age groups suggests a shared understanding of prevailing economic trends.

Overall, the expectations in 1 year ahead inflation rate reflects a period of significant fluctuation followed by a relative stabilization.

Comparing the expected current inflation rate and 1 year ahead inflation rate, it is clear from the figures that 1 year ahead expectations remain a bit higher than current expectations, but expectations for both periods show similar trends over the rounds.

Query 3: Which age group shows closer anticipation to actual inflation rate of the country?

We already examined that there exists difference in inflation expectations along different age groups. Now, to determine which age group shows closer anticipation to actual inflation rate we take an insight into the mean absolute errors, standard deviation of errors of current expected inflation rate.

Age Group	Mean Absolute Error	SD Error	Correlation
Upto25yrs	3.061243	2.141135	0.0732088582
25to30yrs	2.767050	2.476189	0.0512185312
30to35yrs	3.109005	2.972685	-0.0218704411
35to40yrs	3.193558	2.743187	-0.0168189542
40to45yrs	3.778160	3.072675	-0.0628321181
45to50yrs	3.832313	2.680329	-0.0432531421
50to55yrs	3.605081	3.849761	0.0005721176
55to60yrs	2.715587	2.943886	-0.0220183093
Above60yrs	2.653027	1.773485	-0.0127176438

For each age group, Mean Absolute Error is calculated as:

$$S = \frac{1}{45} \sum_{round=1}^{45} \left| mean_expected_current_inflation[round] - actual_inflation[round] \right|$$

Mean expected current inflation for each round, is the mean of expected inflation for an age group, for that round.

SD Error for each age group is nothing but the standard deviation of errors. (i.e., actual_inflation – mean_expected_current_inflation). The correlation column shows the Pearson's Correlation Coefficients between actual inflation rates and expected current inflation rates, for each age group.

The mean error fluctuates across age groups, with higher errors observed in the 40-50 age range (peaking at 3.832313 for 45-50 years). Higher standard deviation (SD) indicates greater variability in errors, meaning more inconsistency in expectations. The highest SD is observed in the 50-55 years group (3.85), implying this group has the most varied errors. The Above 60 years group has the lowest SD (1.77), indicating more stable and consistent errors. The lowest mean absolute error is also seen in the Above 60 years group, suggesting that older individuals might have more accurate expectations or lower deviations from actual inflation.

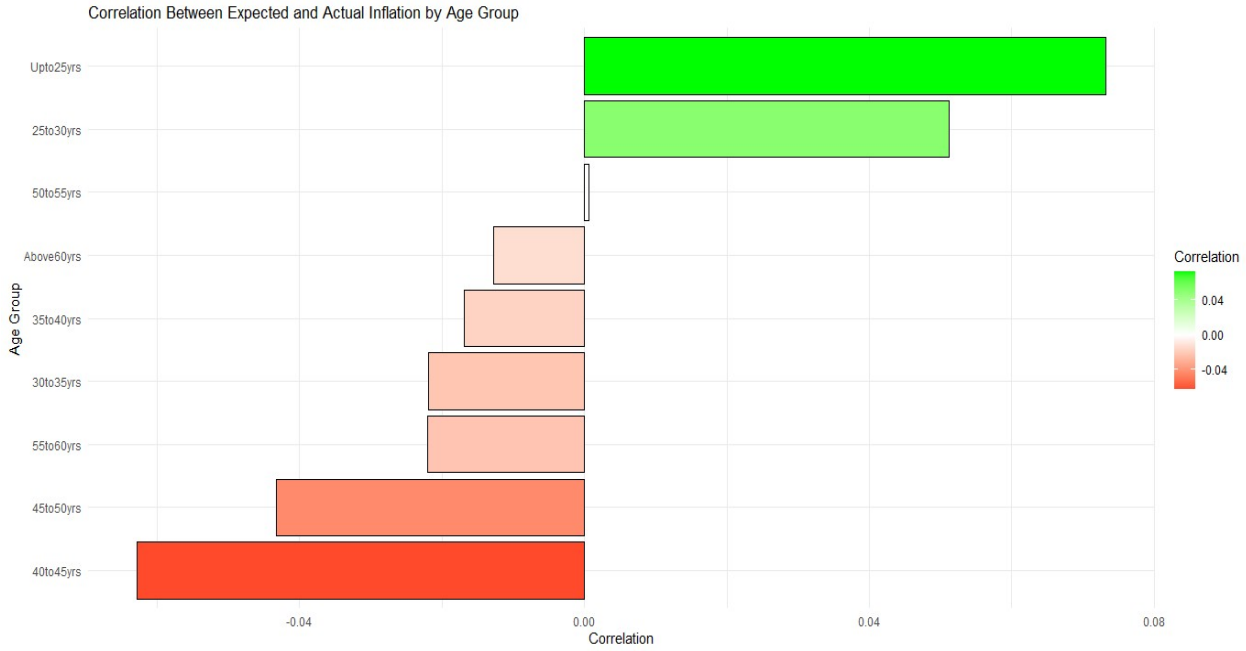


Fig 4

The bar chart, which represents the different Age Groups on its vertical axis and correlation between actual and current expected inflation along the horizontal axis, visually reinforces the correlation values presented in the table. Only the two youngest age groups (Upto 25 years and 25 to 30 years) show the only positive correlations, indicated by the green bars. However, even these positive correlations are extremely weak (0.073 and 0.051 respectively). This suggests that for these younger age groups, there is a slight tendency for their expected inflation to move in the same direction as the actual inflation, but the relationship is not strong. The remaining age groups exhibit negative correlations, denoted by the red bars in the chart. The 40-45 years Age Group shows the strongest negative correlation (-0.0628), though still extremely weak. This indicates a slight inverse relationship where the expected inflation of older individuals tends to move in the opposite direction of the actual inflation possibly shaped by past experiences.

The most striking observation is the overall weakness of the correlations across all age groups. This suggests that expected inflation, as perceived by these age groups, has a very limited linear relationship with the actual inflation rate.

In conclusion, younger individuals (Up to 25 years) have relatively better alignment between expectations and actual inflation. Middle-aged groups (40-50 years) have the highest errors and deviations, suggesting they may have greater difficulty predicting inflation accurately. Older

individuals (Above 60 years) show more stable errors, but their expectations are negatively correlated with actual inflation. The low correlations suggest that using expected inflation from these age groups alone might not be a reliable predictor of actual inflation. Overall, expectation errors exist across all age groups, with no strong correlation to actual inflation, indicating that inflation perception is generally weak across demographics.

Query 4: Analysis of Occupation Distribution Across Cities:

a) What is the overall percentage distribution of the occupation of the population across different cities for the entire survey period (from July 2008 to September 2019)?

b) In what ways has the distribution of occupations across cities changed over the course of the survey period?

a)

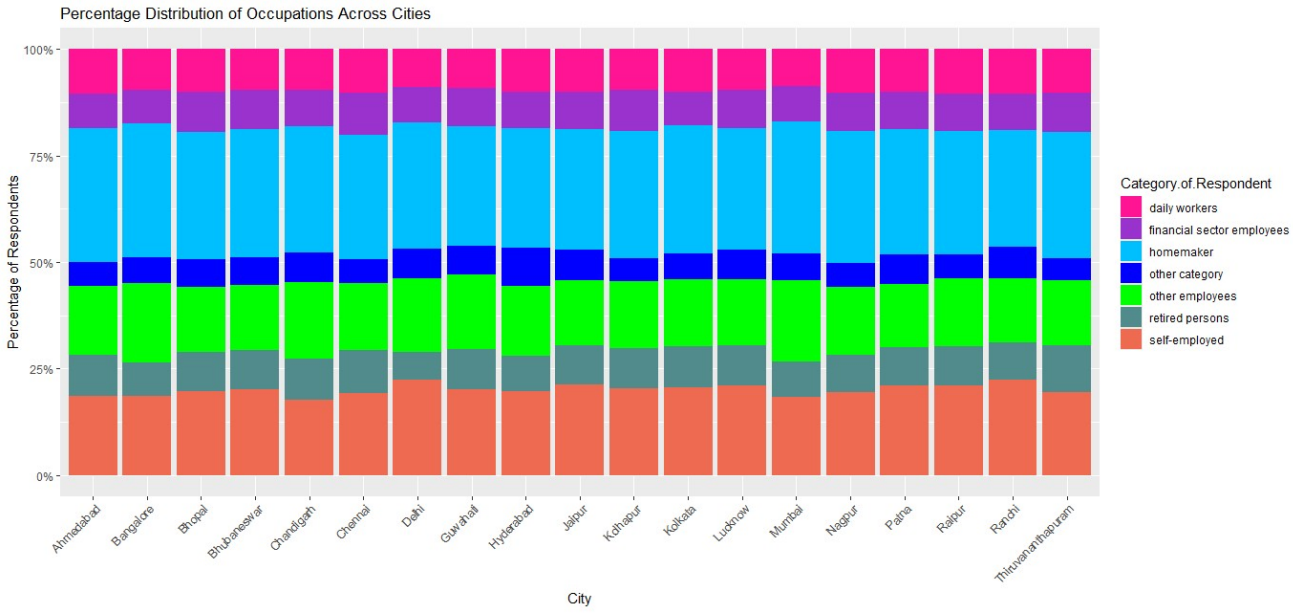


Fig 5

To analyse the occupation distribution of the respondents, from various parts of the country, we have plotted the percentage of frequencies of respondents in a specific occupation for each city in the above Subdivided Bar Diagram. The X-axis plots the Names of the Cities, and Y-axis plots the Percentage of Respondents.

A visual insight into the above diagram shows a dominant trend in the consistent prevalence of homemakers (light blue) across all cities. However, the proportion of homemakers varies, with Ahmedabad exhibiting the highest percentage (31.49%) and Ranchi the lowest (27.32%), indicating presence of regional differences in workforce participation. Self-employment (brown) shows notable fluctuations, peaking in Delhi (22.39%) and reaching its lowest in

Chandigarh (17.75%), potentially reflecting differing economic structures and entrepreneurial opportunities. The other employees (green) category, likely encompassing a broad range of formal sector jobs, is most prominent in Mumbai (19.23%) and least so in Patna (14.77%), hinting at variations in industrial composition and employment patterns.

Daily workers (pink), while a smaller segment, are most concentrated in Ahmedabad (10.51%) and least in Mumbai (8.56%), potentially reflecting the prevalence of informal labour in these cities. The financial sector employee's category, though consistently low, reaches its highest point in Chennai (9.72%) and its lowest in Kolkata (7.89%), suggesting regional variations in the strength of the financial industry. Retired persons are most prevalent in Thiruvananthapuram (10.97%), possibly due to demographic factors or retirement migration, and least in Delhi (6.45%), which might reflect a younger population or different retirement patterns. Finally, the other categories segment, encompassing diverse occupations, is highest in Hyderabad (8.94%) and lowest in Thiruvananthapuram (5.01%), indicating variations in occupational diversity.

In conclusion, the occupation distribution across cities in India highlights significant regional variations in workforce participation, with certain occupations, such as homemakers and self-employment, showing pronounced differences, reflecting the diverse economic structures, demographic factors, and industry strengths of each city.

b) We now consider the occupation distribution of each city, over the 45 Rounds considered. As we are dealing with time series data involving continuous time points, Line Diagram is a suitable method to visualise. Here, for each city, we consider separate Line Diagrams in separate panels. It shows percentage distribution of occupations over time in each city.

The X-axis plots the Round Number and the Y-axis plots the Percentage of Respondents.

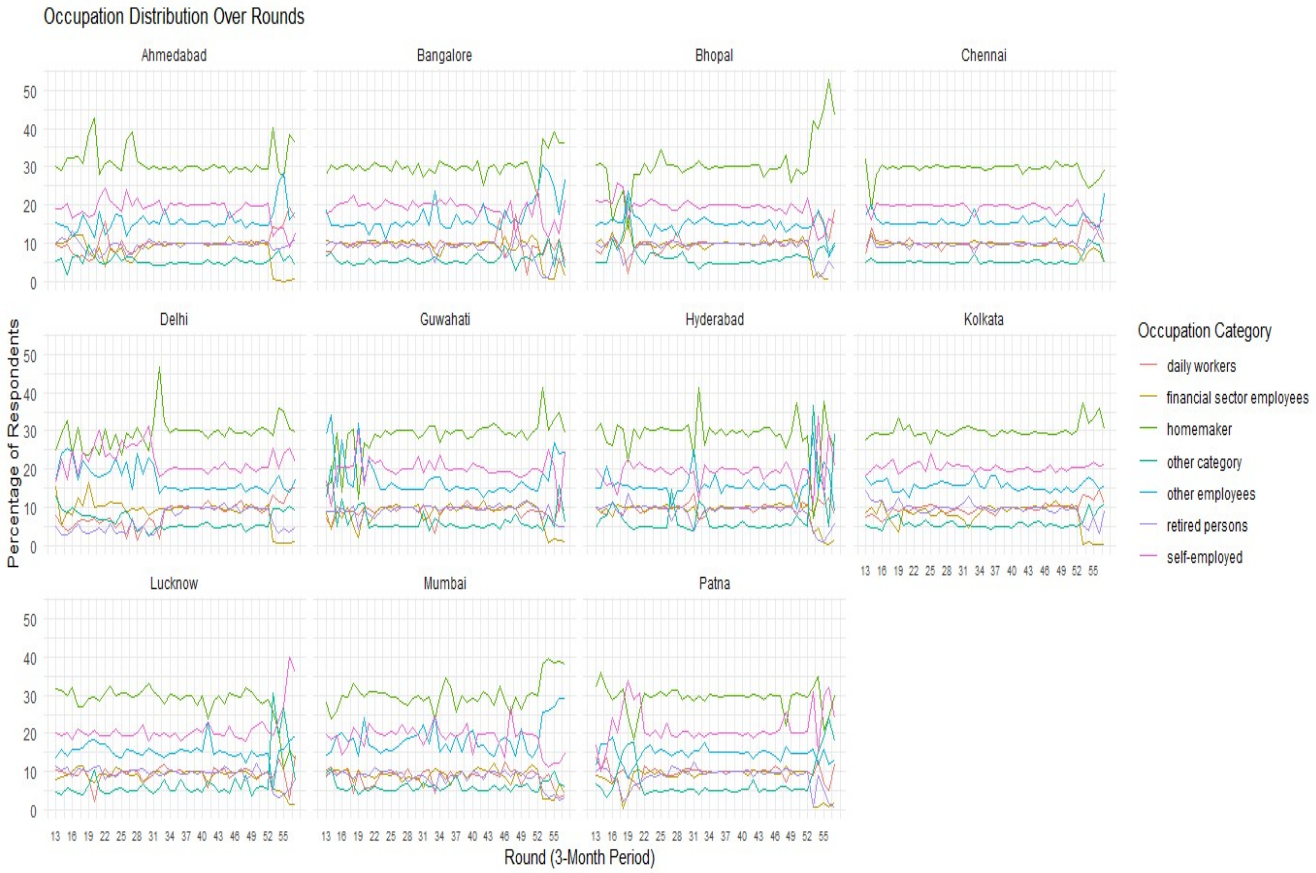


Fig 6

Analysis across occupations: The line graphs effectively reveal dynamic trends over the survey rounds. A consistent pattern across all cities is the dominance of homemakers, consistently exhibiting the highest percentage of respondents, suggesting their enduring presence in the surveyed population. Following homemakers, self-employed individuals and other employees maintain relatively stable, though lower, percentages. Daily workers, financial sector employees, and retired persons demonstrate very low-percentage trajectories, while the other categories group consistently represents the smallest segment throughout the rounds. A widespread trend is the decrease in financial sector employees around round 53, observed in most cities except Chennai, Hyderabad, and Lucknow.

Analysis across cities: All cities overall show high proportion of homemakers, but Delhi exhibits a unique trend where self-employed respondents approach the percentage of homemakers between rounds 18 and 29, suggesting a period of increased entrepreneurial activity. After Round 27, the prevalence of homemakers surpasses other categories. Moreover, Mumbai and Bhopal experience a notable surge in homemaker representation around round 51. Delhi and Hyderabad show a sudden peak in occurrence of homemakers, near Rounds 28-35. In contrast, Lucknow displays a gradual decline in homemakers coupled with a sharp increase in self-employed respondents around round 53, indicating a potential economic or demographic transformation. Mumbai also shows an increase in other employees alongside homemakers from round 52, suggesting a possible expansion in formal employment sectors.

In summary, the analysis highlights distinct regional trends and temporal shifts in occupation distribution, reflecting dynamic changes in economic and demographic factors across cities.

Query 5: How does expected current inflation rate vary among different occupations, and expectations of which occupation aligns most with the actual inflation?

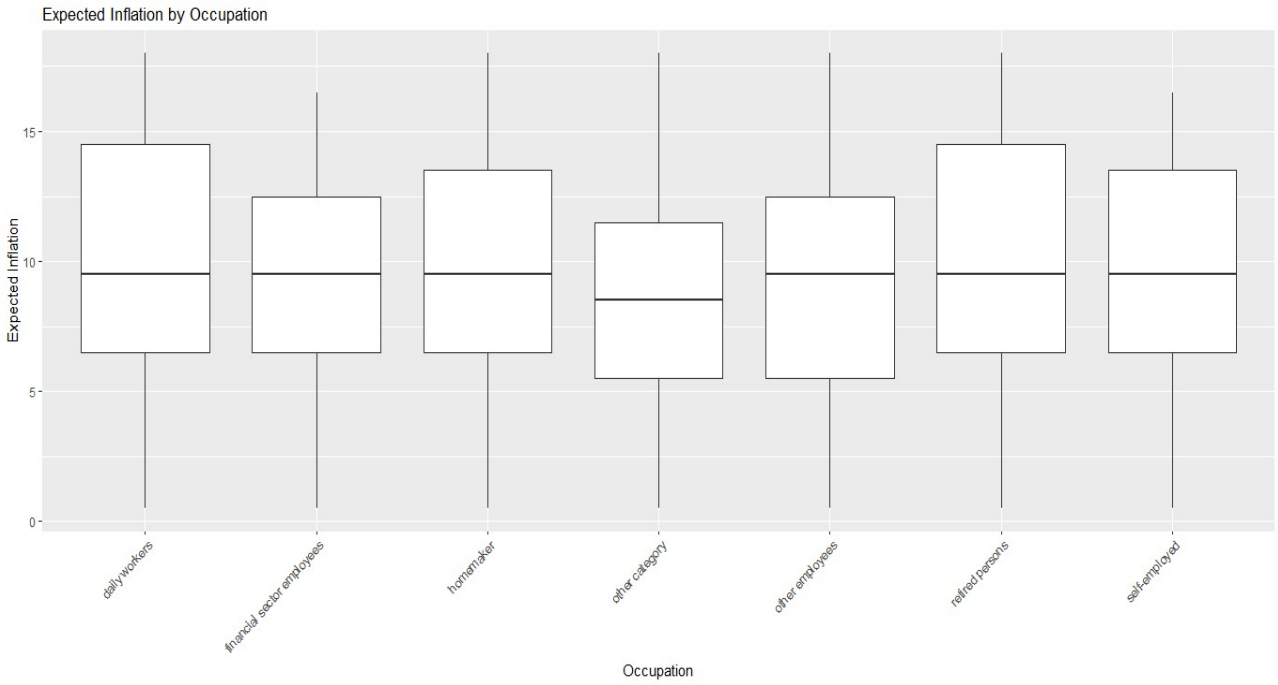


Fig 7

The above boxplot reveals the distribution of expected inflation across seven occupational groups: daily workers, financial sector employees, homemakers, other categories, other employees, retired persons, and self-employed. The above diagram plots the occupational groups along the X-axis and the Y- axis plots the expected current inflation rate.

The median expected inflation remains relatively consistent across all groups, with the median values being around 9%-10%. Furthermore, the spread (IQR) and overall range of expected inflation exhibit similar patterns across occupations, indicating comparable variability in individual inflation perceptions. The whiskers for all the occupations vary from 0% to around 17%. For the financial sector employees and self-employed, the whiskers are a bit shorter, indicating slightly less spread of the data.

Daily workers, homemakers and retired persons show a wider spread in their interquartile range. Among all the occupations, people working in Other Categories seem to expect lower inflation rates than others. Retired persons and daily workers show most variability in their expectations, and have a slightly positively skewed distribution. Financial sector employees

and people in other categories show relatively less variability and a symmetric distribution roughly.

COMPARISON WITH ACTUAL INFLATION:

Occupational Group	Mean Absolute Error	SD Error	Correlation
Homemaker	3.077391	1.619097	-0.12069317
Daily Workers	3.160466	1.589412	-0.14027126
Other Category	3.102798	1.638721	-0.16706633
Other Employee	3.057813	1.641759	0.01595182
Self Employed	3.124671	1.651213	-0.11750118
Retired Persons	3.145557	1.682291	-0.22814324
Financial Sector Employees	3.050925	1.700697	-0.12534407

Based on the provided table, which shows the Absolute Mean Error**,** SD Error**,** and Correlation for various occupation categories, we can draw the following conclusions:

Correlation Analysis:

Here, we considered Pearson's Correlation Coefficient to examine if any category of occupation shows more linearly aligned expectations with actual inflation rate of the country. It is to be noted that the conclusions drawn are with respect of presence of only linear association between expected and actual inflation rates.

A weak correlation is observed between expected and actual inflation across all occupational groups. This suggests a significant misalignment between individual perceptions and the actual inflation rates. Specifically, Retired Persons exhibit the lowest correlation (-0.2281), indicating their expectations are the least aligned with reality. Daily workers, homemakers, and self-employed groups show weak negative correlations. While respondents under 'other employees' category are the only ones that show a slightly positive correlation, it is almost zero (0.0156). Even Financial Sector Employees, despite having the lowest mean error, display a weak negative correlation (-0.1253).

Analysis of Mean Absolute Error and S.D of Error:

Further analysis of the table reveals insights into the accuracy and variability of inflation predictions. Financial Sector Employees demonstrate the lowest mean error, indicating the closest average predictions, but exhibit the highest standard deviation of error, suggesting greater variability in their predictions. Conversely, Daily Workers show a higher mean error but lower standard deviation, indicating more precise but less accurate predictions.

Any occupation neither shows a strong positive correlation with actual inflation, nor shows significantly consistent predictions with lower mean absolute errors compared to others, which means that inflation expectations are, to varying degrees, misaligned with the actual inflation rate across all occupation groups.

Overall, the data suggests that no occupation category has perfect alignment with actual inflation, and there are varying levels of accuracy and variability. The Financial Sector Employees show the lowest mean errors but also the highest variability, while Daily Workers and Homemakers tend to have higher mean errors but less variation in expectations. Retired Persons appear to be the least accurate group, with lowest correlation between actual and expected current inflation, relatively high mean absolute error and high standard deviation of error.

Query 6: How are respondents' expectations of price changes for various commodities related with each other, and what notable patterns can be identified in the relationships between these expectations over the next three months?

The data on expectations of price changes for next 3 month and 1 year on different categories of commodity are ordinal, given in 5 categories, which we code as follows:

a) Decline in price: -0.5
b) No change in price: 0
c) Price increase less than current rate: 0.5
d) Price increase similar to current rate: 1
e) Price increase more than current rate: 1.5

Since the data is not numerical, we cannot go for Pearson's Product Moment Correlation Coefficient. We calculate Spearman's Rank Correlation to examine the relationships between respondents' expectations of price changes across different commodity categories (General, Food, Non-food, House Durables, Housing, and Services). This method helps us identify and measure the strength and direction of monotonic relationships between the variables, without assuming a linear relationship.

By identifying these correlations, we can gain insights into how interconnected the respondents' views are across different sectors.

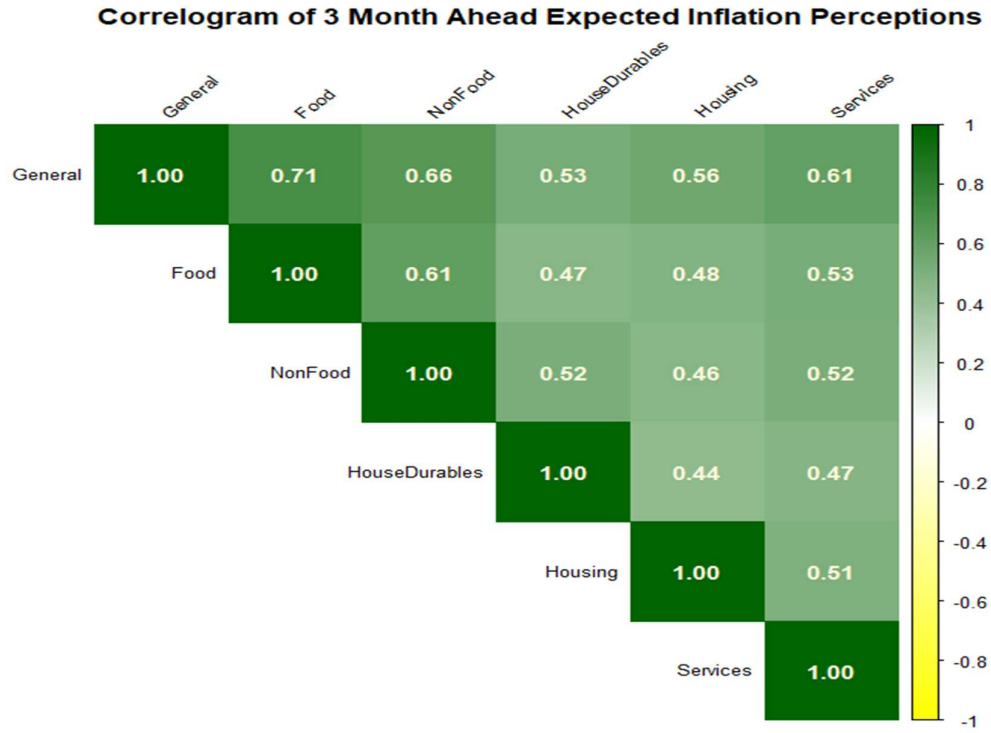


Fig 8

The correlogram provides a visual representation of the Spearman's Rank Correlation, across different categories of commodities depicting the relationships between respondents' expectations for price changes across the six categories. The colour intensity corresponds to the strength of the correlation with darker green hues indicating stronger positive correlations, white indicating zero correlation and yellow for negative correlation. The actual numerical values of the correlation coefficients are displayed within the cells.

It is to be noted that the darkest green cells, with Spearman's rank correlation as 1 are the correlation of the variables with themselves, so the data on both are same and the correlation is 1. The darker green cells, representing the strongest positive correlations, are observed between general commodities and food products (0.71) as well as between non-food products and general price level, indicating a strong alignment in respondents' expectations for general price changes and food and also between general and non-food price changes. The correlogram also shows moderately strong positive correlations (lighter green cells) between price change of food and non-food products, between general price levels and service, general and housing, general and house durables.

It reveals moderate positive correlations (even lighter green cells) between house durables and food products (0.47), house durables and housing (0.44), and housing and services (0.51).

In conclusion, in the correlogram, the predominantly green cells indicate a general trend where respondents expect price changes in different sectors to move in the same direction, reflecting the interconnectedness of the economy. The correlogram provides a valuable tool for understanding how price expectations in one sector may influence respondents' views in others, with the General category serving as a key indicator of overall price expectations. The visual

representation highlights that while there is an overall positive correlation, the strength of the correlation differs among the different categories.

Query 7: (a)What proportion of respondents expect various levels of price increases over the next three months and the upcoming year for different categories of commodities, such as general, goods, food, non-food items, housing, house durables, and services?

(b) Does there exist any relation between zones and the different levels of expected price, for each category of commodity?

(a) The data provided records different levels of 3-month ahead and 1-year ahead expectations of price level, where the data on expectation is ordinal, categorised into 5 levels: Decline in Price, No change in Price, Price increase less than current rate, Price increase similar to current rate and Price increase more than current rate.

Expected Commodity Price Changes for next 3 Months:

For this, we calculated the percentage of respondents opting for different levels of price expectations in each considered sector of the economy. To represent this diagrammatically, Subdivided Bar Diagram is used to show 3 months ahead expected price level of different commodities. In X-axis different categories of commodities (nominal) are plotted and in Y-axis the percentage of respondents (ratio), opting for different levels of price change is plotted. For this, we calculated the percentage of respondents opting for different levels of price expectations in each considered sector of the economy.

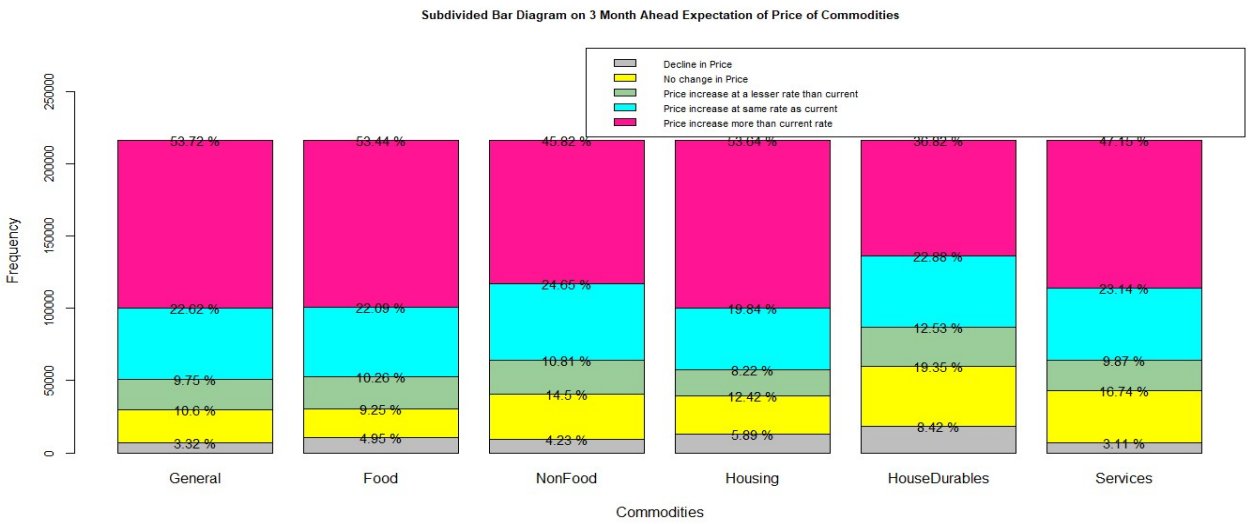


Fig 9

The Subdivided Bar Diagram reveals that respondents predominantly expect commodity prices to increase faster than the current rate over the next 3 months.Across all categories (general, food, non-food, housing, household durables, and services), a significant portion of respondents anticipates accelerated price increases. Housing exhibits the highest expectation of rapid price increases (53.84%), indicating significant concerns about housing affordability in the short term.On the other hand household durables show the lowest expectation of accelerated price increases (38.82%), suggesting perceived short-term price stability. The expectation of price declines is consistently low across all categories, indicating a lack of optimism about short-term price reduction.

Expected Commodity Price Changes for Next 1 Year:

Similarly to study expected price levels in next 1 year, we plot a subdivided bar diagram.

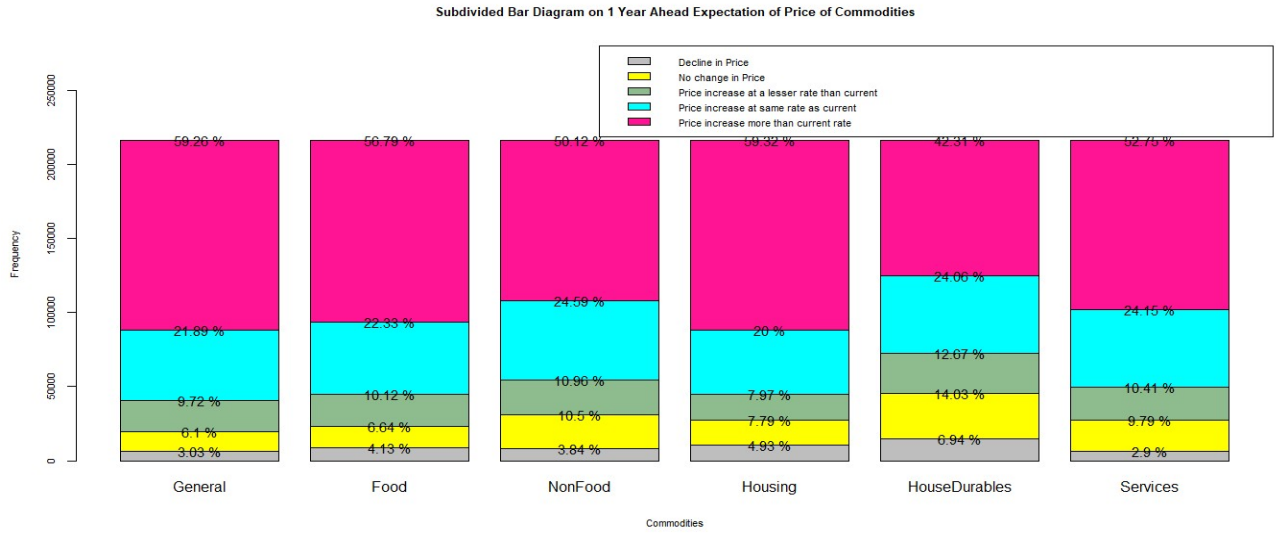


Fig 10

Expectations of price level for the next 1 year show a similar trend of anticipated price increases, with some variations in magnitude. Respondents continue to expect prices to increase faster than the current rate across all catagories. Housing again exhibits the highest expectation of rapid price increases, with an even higher percentage (59.32%) compared to the 3-month forecast. Household durables again show the lowest expectation of accelerated price increases (42.31%). Similar to the 3-month outlook, the expectation of price declines remains consistently low.

Conclusion: (Comparing 3 Months and 1 Year)

Overall, respondents consistently anticipate accelerated price increases for all commodity categories over both the next 3 months and 1 year. While the relative ranking of commodity categories in terms of price change expectations remains largely consistent between the two timeframes, the percentage of respondents expecting higher inflation generally increases when looking at the 1-year horizon. This suggests a strengthening belief in persistent inflationary pressures over the longer term. The percentage of respondents for each category of commodities decreases as the expected price level shifts from higher to lower levels.

(b) Now, for each category of commodity, we separately prepare a contingency table, with one attribute as Zone (Nominal) and another as Price Level (Ordinal).

Here our objective is to know how respondents' perception about price rate differs among different zones, for each category of commodity. The attribute respondents' perception about price is an ordinal attribute. The data is collected across 4 Zones, viz., "East", "North", "South", "West".

Let A be the attribute denoting the respondents' perception on expected price level and B be the attribute denoting the 4 zones.

Let us assume fij be the number of people anticipating ith level of price change from jth zone of India.

Here, i=1,2,3,4,5 corresponds to 'Decline in Price', 'No change in Price', 'Price increase less than current rate', 'Price increase similar to current rate' and 'Price increase more than current rate' respectively.

j=1,2,3,4 corresponds to 'East', 'North', 'South', and 'West' zone respectively.

Now, fi0 be the total number of people anticipating ith level of price change and f0j be the total number of people from jth zone. (i=1,2,3,4,5 j=1,2,3,4). We define,

$$\chi^2_{AB} = n(\sum_{i=1}^{5}\sum_{j=1}^{4}\frac{f_{ij}^2}{f_{i0}\times f_{0j}} - 1)$$

Now as χ^2_{AB} involves n, it is not a bounded measure. To overcome this limitation, we compute C_{AB}

$C_{AB}=\sqrt{\frac{\chi^2_{AB}}{n+\chi^2_{AB}}}$ $0\leq C_{AB}<1$

We have shown the detailed steps to obtain **C_{AB} for the expectation on general price level** against 3-month ahead levels of price expectations.

Contingency Table (Zone vs Expected 3-month ahead price level) for General Commodities:

Price level / Zone	Decline in prices	No change in prices	Price increase less than current rate	Price increase similar to current rate	Price increase more than current rate
East	1568	4570	4890	11831	20496
North	2015	6789	5648	14304	31404
South	1443	4543	4420	9022	33420
West	2143	7005	6118	13736	30794

Similarly, we obtain contingency tables for each category of commodities, for 3-month as well as 1 year ahead expectations. Now, we obtain C_{AB} from each of the contingency tables, and get the following results:

C_{AB} values: Zone vs Level of price:

CATEGORY	C_{AB} VALUE (3-month)	C_{AB} VALUE (1- year)
General	0.1187	0.09744
Food	0.107577	0.088675
Non-food	0.14116	0.102356
Housing	0.114127	0.11472
House durables	0.1058	0.0956
Services	0.133743	0.11068

C_{AB} value ranges from 0 to 1, with 0 meaning no association between the attributes, and 1 meaning perfect association. Here, for 3-month ahead expectations, we see that all the values are concentrated around 0.10-0.15, showing very less variation. The strength of association between zone and expected price level is very low, and does not differ across commodity categories. Hence, we can conclude that zones have very less influence on expectations in 3-month ahead price levels in all the categories of commodity considered. Among these low C_{AB} values, Non-food items has the highest and House Durables has the lowest C_{AB} value.

Again, for 1 year ahead expectations, we see that all the values are concentrated around 0.09-0.11, showing lesser variation. The strength of association between Zone and 1 year ahead expected price level is very low, and does not differ across commodity categories. Hence, we can conclude that Zones have very less influence on expectations in 1 year ahead price levels in all the categories of commodity considered. Among these low C_{AB} values, housing has the highest and House durables has the lowest C_{AB} value.

In both 1 year ahead and 3- month ahead expectations, House Durables has the lowest association with zones. The 1-year ahead expectations show lesser C_{AB} value than the 3-month expectations. Hence, the association between zones expected price level is very weak, and weakens as we move from 3-month ahead to 1 year ahead expectations.

Thus, we see that people from all over the country mostly expect higher price levels in the future (mostly increase in price more than current rate), which does not depend on zonal classification.

Query 8: Does gender influence expectations regarding the current inflation rate?

To get an initital visual idea about the expectations of male and Female respondents on current inflation, we calculate mean of expected inflation for each round, separately for males and female, and then plot the means over the rounds, through a Line Diagram. On the same graph, we also plot the actual inflation rate of the country. X-axis shows the Round Numbers and Y-axis shows the mean of expected inflation rate with respect to each round. It can be seen that the expected inflation for both the gender are mostly aligned throught all the rounds, but none aligns with the actual inflation rate of the country. To study this alignment in expectation between the two genders, we consider examination though Point Biserial Cirrelation Coefficient.

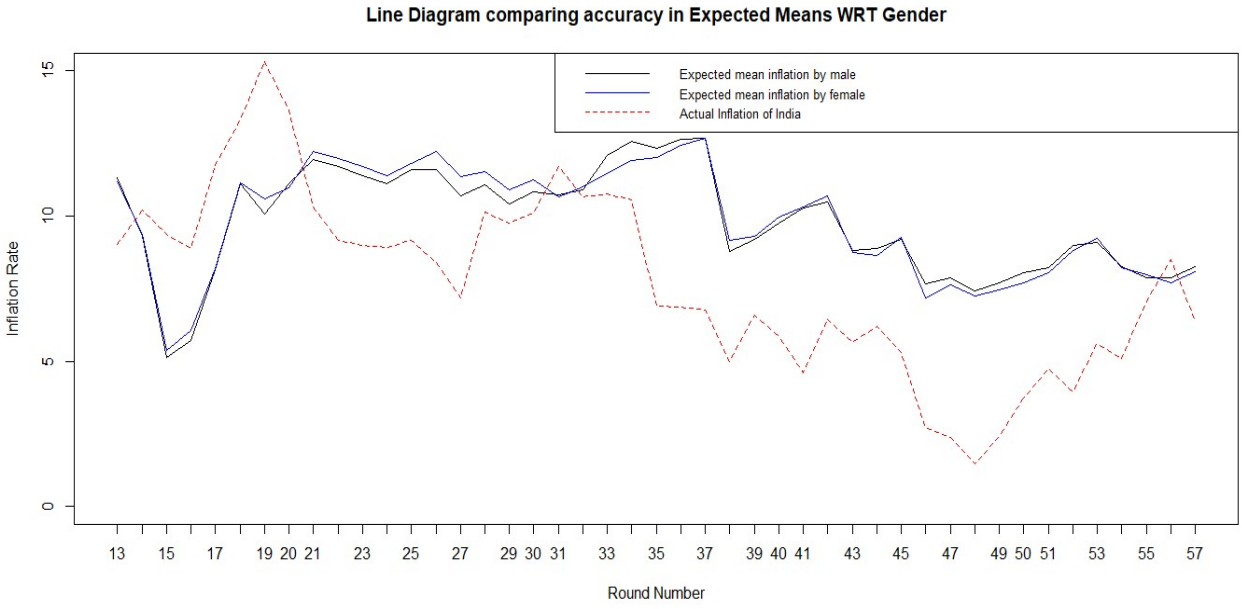


Fig 11

In this analysis, we aim to examine the relationship between gender (a binary variable, coded as 0 for male and 1 for female) and the expected mean inflation rate (a continuous variable). Point-biserial correlation is an ideal statistical method for this scenario as it specifically assesses the association between a dichotomous variable and a continuous variable. It assesses whether there's a statistically significant association between the two.

The point biserial coefficient is given by,

$$r_{pb} = \frac{M_1 - M_0}{S_{n-1}} \times \sqrt{\frac{n_1 \cdot n_0}{n \cdot n}}$$

, where M_1 and M_0 are means of mean of expected current inflation rate by females and male respectively and n_1, n_0 are number of female and male respondents respectively.

We assume that current expected inflation rate is approximately normally distributed. We consider a test of this Biserial Co-efficient, with,

Null hypothesis, H_0: There is no difference in expected current inflation rates between males and females (i.e., $r_{pb} = 0$).

Alternate Hypothesis, H_1: There is a difference in expected current inflation rates between male and female respondents (i.e., $r_{pb} \neq 0$).

Here the suitable Test Statistic is given by:

$$T = r_{pb} \times \sqrt{\frac{(n_1 + n_0 - 2)}{1 - r_{pb}^2}}$$

Here, under H_0, T follows a t distribution with n_0+n_1-2 degrees of freedom.

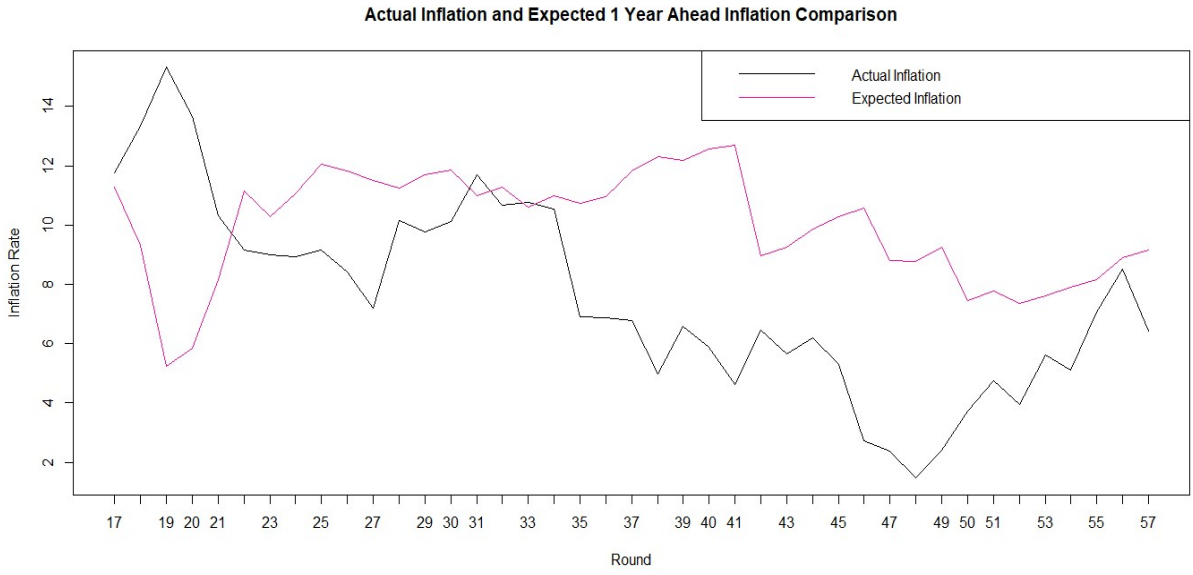
After calculating the value of test statistic, we get that the test yielded a p-value of 0.279. This p-value is greater than the standard significance level of 0.05. Consequently, we fail to reject the null hypothesis at 5% level of significance.

This suggests that the observed differences in expected inflation rates between males and females are likely due to random variation rather than a genuine effect of gender. While the line graph might show some visual differences between the expected inflation rates of males and females across survey rounds, the statistical analysis indicates that these differences are not significant enough to reject the notion that males and females, on average, hold similar expectations regarding inflation.

Here it is important to note that this conclusion is based on the assumption that the expected inflation rates are approximately normally distributed. Furthermore, the analysis only examines a linear relationship between gender and expected inflation.

Query 9: How closely do expected inflation rates for one year ahead align with the actual inflation rate observed one year later?

The objective here is to study if the 1 year ahead inflation expectations align with the actual inflation rate at the period for which anticipation is done (i.e., after 1 year or after 4 rounds). First, we shortened the mean current expected inflation vector, to contain expected inflations of round 13 to 53, and the actual inflation to contain from round 17 to 57. Here, expected 1 year ahead inflation rate of round 13 corresponds to actual inflation rate of round 17, expected of round 14 corresponds to actual inflation of round 18 and so on.



We plot round on X axis, and plot the expected 1 year ahead inflation for round x on round x+4 in the diagram, as rounds are 3-month periods. A visual insight into the graph shows presence of significant difference in the expected 1 year ahead inflation and the actual inflation after 1 year.

We assume both the expected 1 year ahead inflation and actual inflation follow Normal Distribution, with unknown means, variances, and correlation coefficient.

X_1: Expected 1 year ahead inflation

X_2: Actual inflation of the country after 1 year

Then $(X_1, X_2) \sim BN(\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, \rho)$

Let (X_{11}, X_{21}), $(X_{12}, X_{22})\ldots (X_{1n}, X_{2n})$, where n=41 (number of rounds considered).

Let X_i denote the mean of observed values on X_i and s_i^2 denote the sample variances of the same respectively. (i=1, 2)

To test:

- **Null Hypothesis (H_0)**: There is no difference between expected 1 year ahead inflation and actual inflation after 1 year. (i.e., H0: $\mu 1 - \mu 2 = \mu 0$).
- **Alternative Hypothesis (H_1)**: There is a significant difference between expected 1 year ahead inflation and actual inflation after 1 year. (i.e., H_1: $\mu_1 - \mu_2 \neq \mu 0$)

, where $\mu 0 = 0$.

Let $Z = X_1 - X_2$. Then, $Z \sim N (\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2 - 2.\rho.\sigma_1.\sigma_2)$. Let $\bar{Z}$ and s_z^2 respectively denote the sample mean and variance of Z.

Here, we perform a paired t- test. We consider the level of significance as 0.05. The test statistic is given by,

$$T = \frac{\sqrt{n}\,(\bar{Z} - \mu_0)}{s_z}$$, which under H_0 follows a t – distribution with n-1 degrees of freedom,

The observed value of T from our sample is obtained as 5.8867. The p-value of the test is 0.000000682, which is much less than 0.05. So, we reject the null hypothesis.

The 95% Confidence Interval is given by [2.383, 4.876], meaning the value of $\mu 0$ lies in this range with 95% probability. But this interval does not contain 0, reinforcing that there is high probability of the 1 year expected inflation being greater than the actual on an average.

Hence, expected 1 year ahead expectations overestimate the actual price levels after one year.

Conclusion:

This analysis of secondary data from the Inflation Expectations Survey of Households (IESH) highlights the complex dynamics of inflation expectations across different demographics and regions in India from July 2008 to September 2019. The study reveals several important patterns and insights, especially in relation to how expectations vary by geography, occupation, and age, and how well these expectations align with actual inflation trends.

The study indicates that while demographic and regional factors influence inflation expectations to some extent, they do not lead to accurate forecasting. Inflation expectations across all groups are diverse, often overestimated, and poorly correlated with actual inflation. Expectation of individuals are not at all at par with the reality, indicating lack of knowledge on economic aspects and financial structure among the majority of the population. Overall, the study highlights the gap between public perception and economic reality.